

Weekly report: Theoretical framework for EIT in NV centers

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Question

Why is it possible to realize room-temperature EIT in alkali atoms, whereas it is difficult in NV centers?

Hypothesis

In NV centers, higher temperature requires a larger control-field Rabi frequency for EIT, which makes the system more likely to deviate from an ideal three-level configuration.

Evidence

(1) The dominant thermal decoherence mechanisms differ fundamentally between alkali vapor and solid-state NV centers: Doppler broadening($\propto \sqrt{T}$) is central in alkali atoms, whereas phonon-induced relaxation and dephasing($\propto T^5$) dominate in NV centers.(図 1)

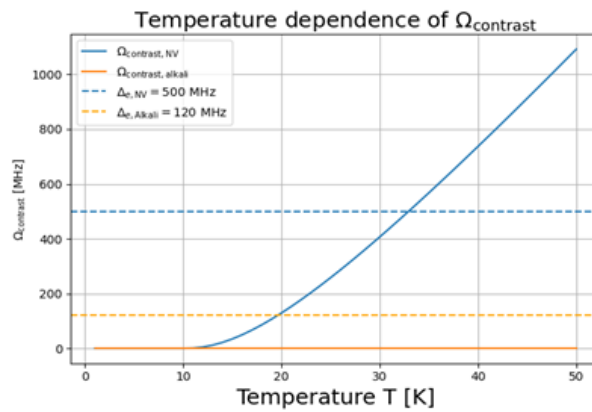


図 1: Comparison of Ω_c dependence of EIT contrast in NV centers and alkali atoms at different temperatures.

(2) In NV centers, the temperature-dependent increase in decoherence raises the threshold

control-field strength for EIT, while simultaneously enhancing the risk of unwanted coupling to nearby excited states.(图 2)

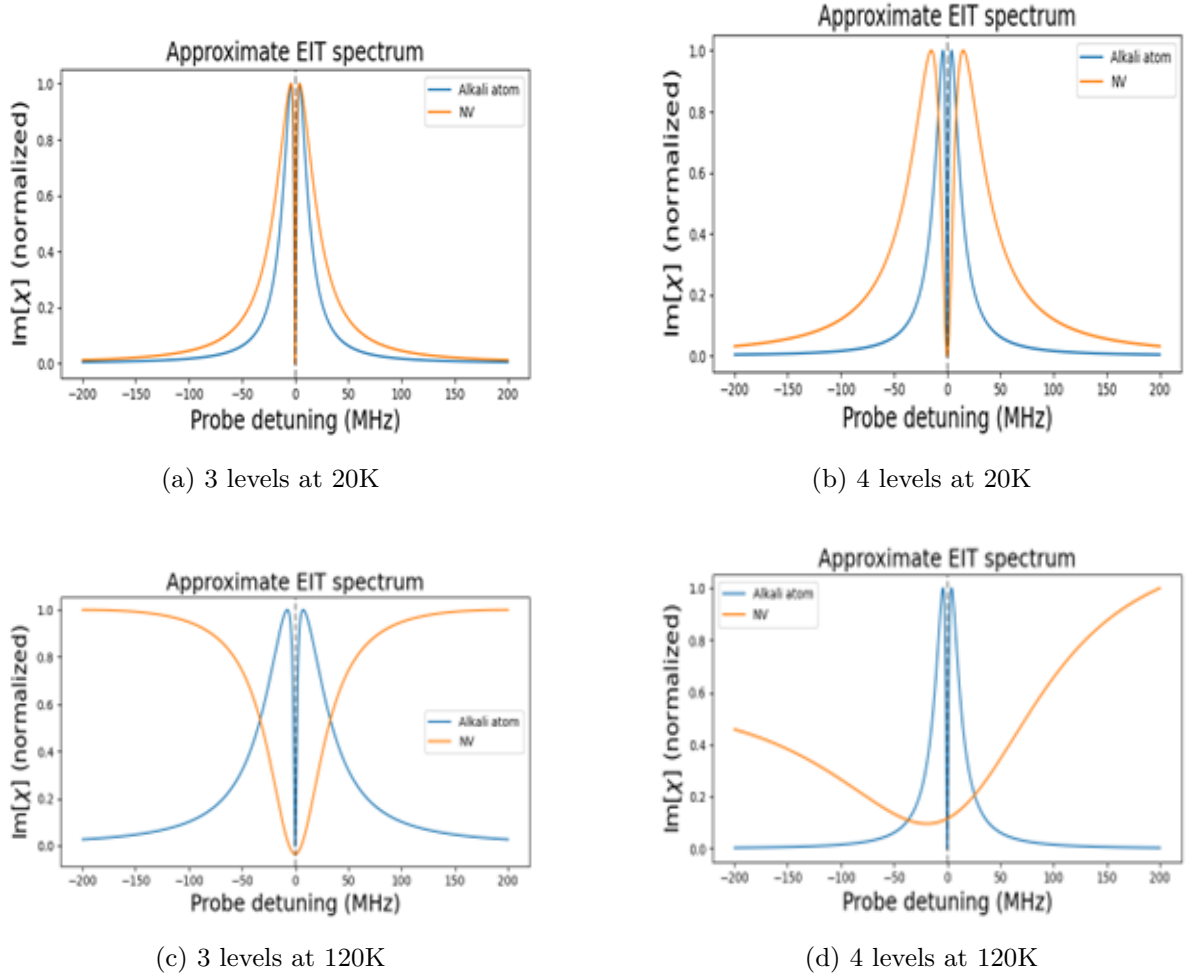


图 2: Comparison of EIT behavior in NV centers and alkali atoms at different temperatures and level structures.

Gap

(1) A quantitative numerical study is needed in which a theoretical framework analogous to that used for alkali-atom EIT is reformulated for NV centers, in order to confirm under what conditions EIT collapses.

(2) The realizable temperature range of NV-EIT is expected to be determined by the energy separation Δ_e from nearby excited states outside the three-level manifold

Next step

Since Δ_e is determined by magnetic field and crystal strain, the next step is to analyze how large Δ_e can realistically become and, based on that, estimate the theoretical upper temperature limit for realizing NV-based EIT.